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Practical 4

Aim: To develop an IoT application for Motion detection using Cisco Packet Tracer

Theory:

Motion Sensor: Motion sensors have a lot of impact in security situations, since they can detect the presence of intruders or sudden changes in the environment.

Motion sensors are small, connected devices that are used to detect and measure movement within a specific area. These sensors are very popular and an essential component of the IoT.

Motion sensors can be used in a variety of applications:

- Home automation.
- Security.
- Healthcare.
- Mobility.

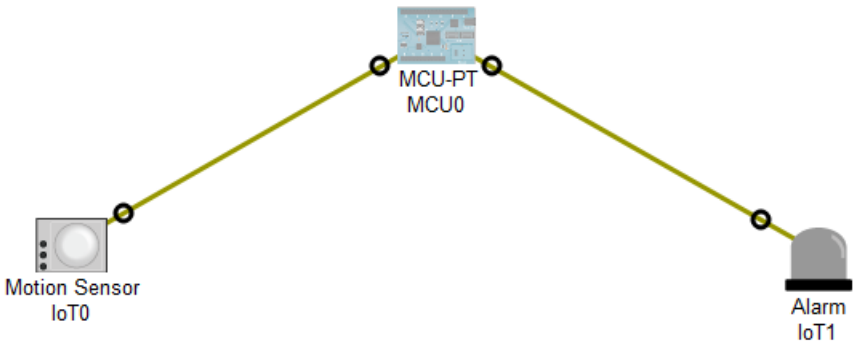
Among the benefits of using motion detector devices, we can find:

- Security improvement by getting alerts when a person approaches a door or window of your house.
- Show when there is a person in a restricted area of the company or facility.
- Save energy through controlling the time a light bulb is turned on in a room, outdoor spaces, or office areas.

Requirement:

- Motion sensor
- MicroController Unit
- Alarm

Diagram:



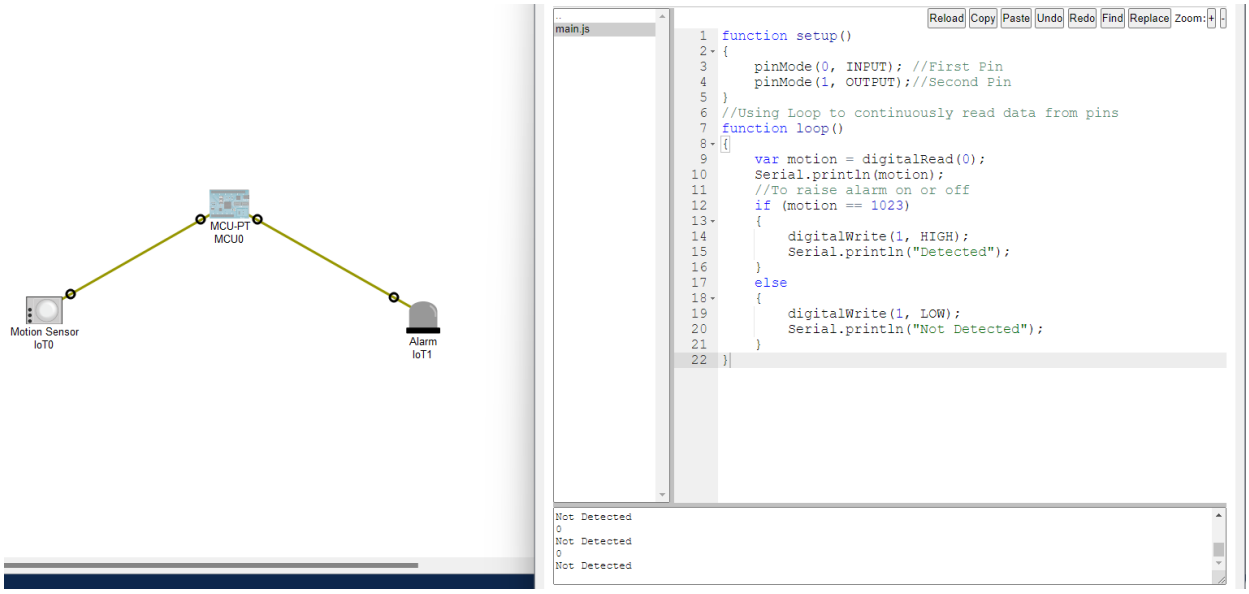
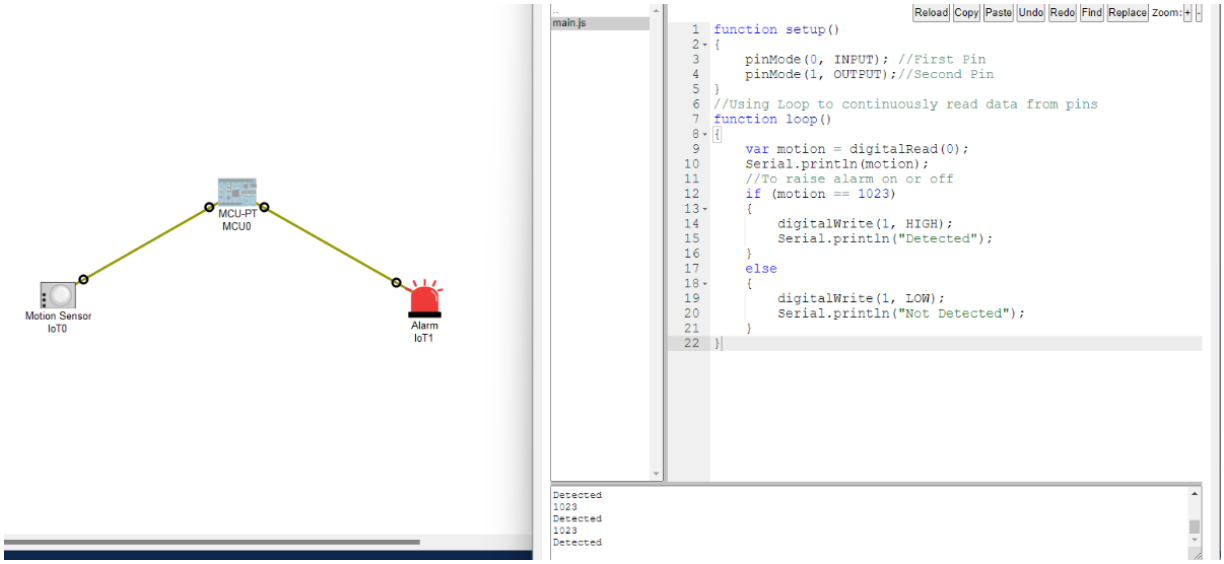
Coding:

To write the code we have click on MicroController Unit. Then go to Programming tab. Create a new JavaScript file and write the code.

```
function setup()
{
        pinMode(0, INPUT); //First Pin
        pinMode(1, OUTPUT);//Second Pin
}
//Using Loop to continuously read data from pins
function loop()
{
        var motion = digitalRead(0);
        Serial.println(motion);
        //To raise alarm on or off
        if (motion == 1023)
        {
            digitalWrite(1, HIGH);
            Serial.println("Detected");
        }
        else
        {
            digitalWrite(1, LOW);
            Serial.println("Not Detected");
        }
}
```

Output:

When motion is not detected alarm will be off. In console it will show ‘0’ and ‘Not Detected’.



When motion is detected will alarm will turn on. In console it will show ‘1023’ and ‘Detected’. To detect motion just hover the mouse cursor over motion sensor.